AI-Powered Revenue Driver Tree — Rebuilt Defensible Model

Observed evidence separated from derived proxies and scenario-only layers

Confidence score: **65 / 100** (rubric-backed, not arbitrary).

Headline Findings

• Observed streaming MRR: 12026.00

• Observed streaming ARR: 144312.00

• Retention weighted AUC: 0.716

• Retention weighted PR-AUC: 0.627

• Retention weighted Brier score: 0.225

• Confidence score: 65 / 100

What Changed

• Static retention was replaced with contract-specific hazard curves and cohort handling.

• Monthly contract artifact was stabilized using a proxy curve borrowed from non-monthly behavior and explicitly labeled as such.

• MRR logic was corrected: streaming MRR is directly observed; SaaS MRR uses a contract-rate-card proxy instead of backward-looking spend averaging.

• Dataset mixing was stopped. Streaming and churn data are no longer fused as one customer system.

• Sensitivity analysis now perturbs upstream drivers and contract mix rather than mechanically multiplying outputs.

• Confidence intervals were relabeled as empirical 80% bands.

• Anomaly detection was added for terminal trajectories, CLV<CAC, constant fabricated outputs, and economically nonsensical results.

• Bookings ACV/TCV now use actual contract-term mix instead of assuming all contracts are annual.

• Scenario layers are explicitly isolated and warning-labeled.

Observed vs Derived vs Scenario

• Observed: streaming acquisition history, streaming active-member MRR, contract mix from churn data, and churn-label-based retention segmentation.

• Derived/proxy: SaaS contractual MRR proxy, ACV proxy, TCV proxy, cohort hazard curves, confidence score rubric.

• Scenario-only: reactivation, upgrade, seat expansion, retention seasonality modifier, CAC proxy, and portfolio transfer forecast.

Retention Validation

contract_type	actual_churn_rate	mean_predicted_churn	auc	pr_auc	brier_score	calibration_ratio
Monthly	0.516	0.509	0.735	0.696	0.221	0.986
Quarterly	0.440	0.460	0.711	0.585	0.225	1.044
Annual	0.462	0.461	0.701	0.596	0.228	0.997

Contract	ECE	Max gap	Pass

Annual	0.111	0.242	False
Monthly	0.102	0.316	False
Quarterly	0.117	0.281	False

Confidence Scorecard

Criterion	Points	Pass	Awarded
Observed streaming acquisition/MRR engine	20	True	20
SaaS retention discrimination gate (AUC weighted >= 0.70)	15	True	15
SaaS retention calibration gate pass rate >= 0.67 of contract types	20	False	0
Monthly-contract artifact stabilized and explicitly proxied	5	True	5
Dataset row-level mixing removed	10	True	10
Scenario layers isolated and warning-labeled	5	True	5
Upstream sensitivity analysis implemented	10	True	10
No critical anomaly alerts in headline engines	10	False	0
Domain transfer validated for portfolio application	5	False	0

Important Warnings

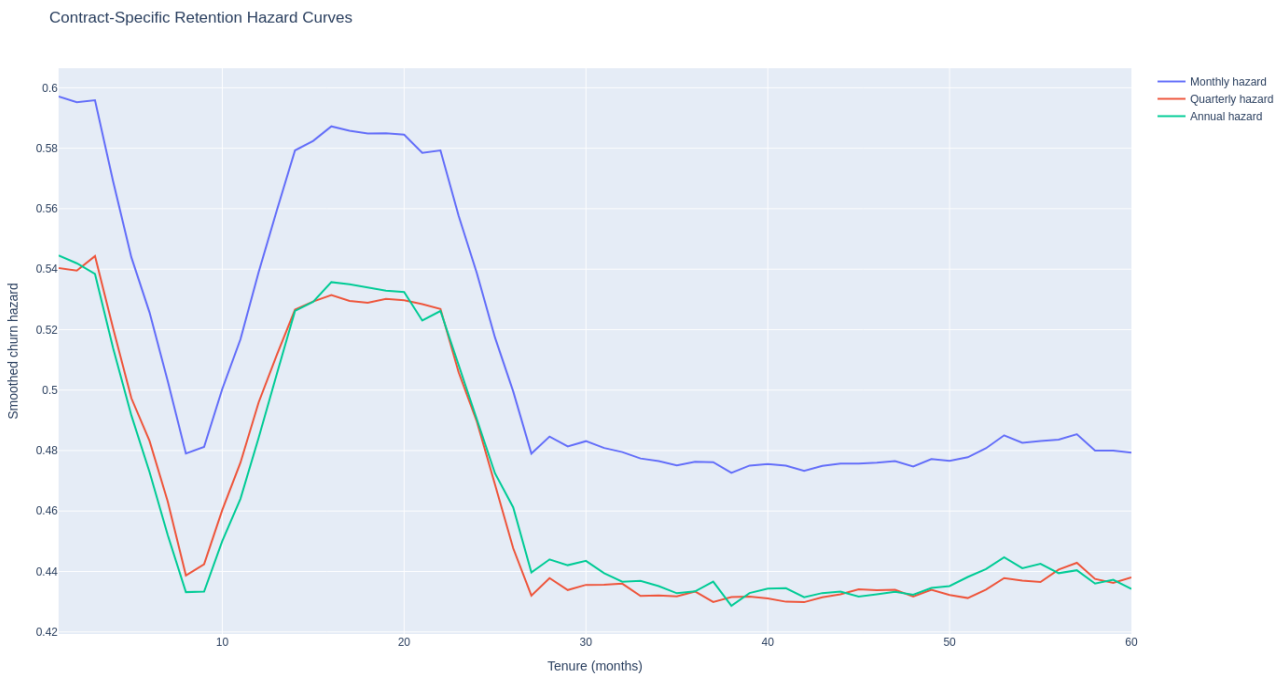
• Monthly retention is proxy-stabilized, not directly learned from monthly training observations.

• Portfolio scenario outputs are appendix-grade transfer scenarios, not headline forecasts.

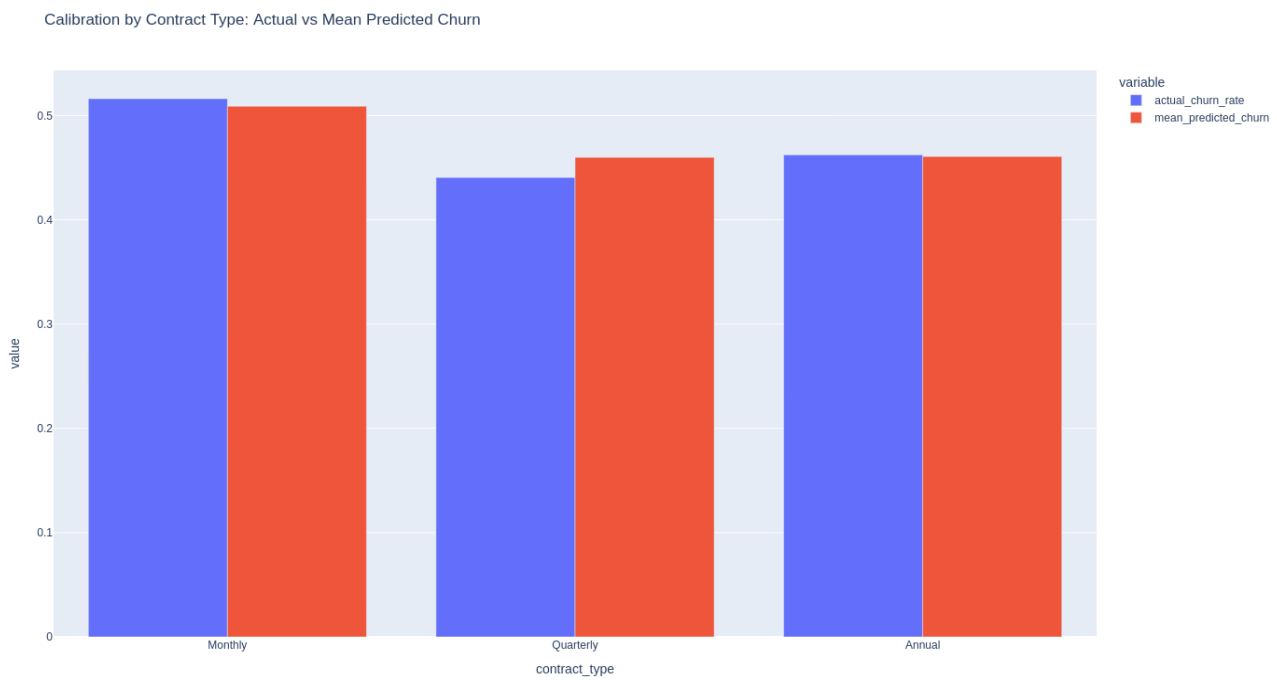
• Critical anomaly alerts are firing; these should be treated as warnings, not as deployable planning outputs.

Appendix: Scenario and Diagnostic Charts

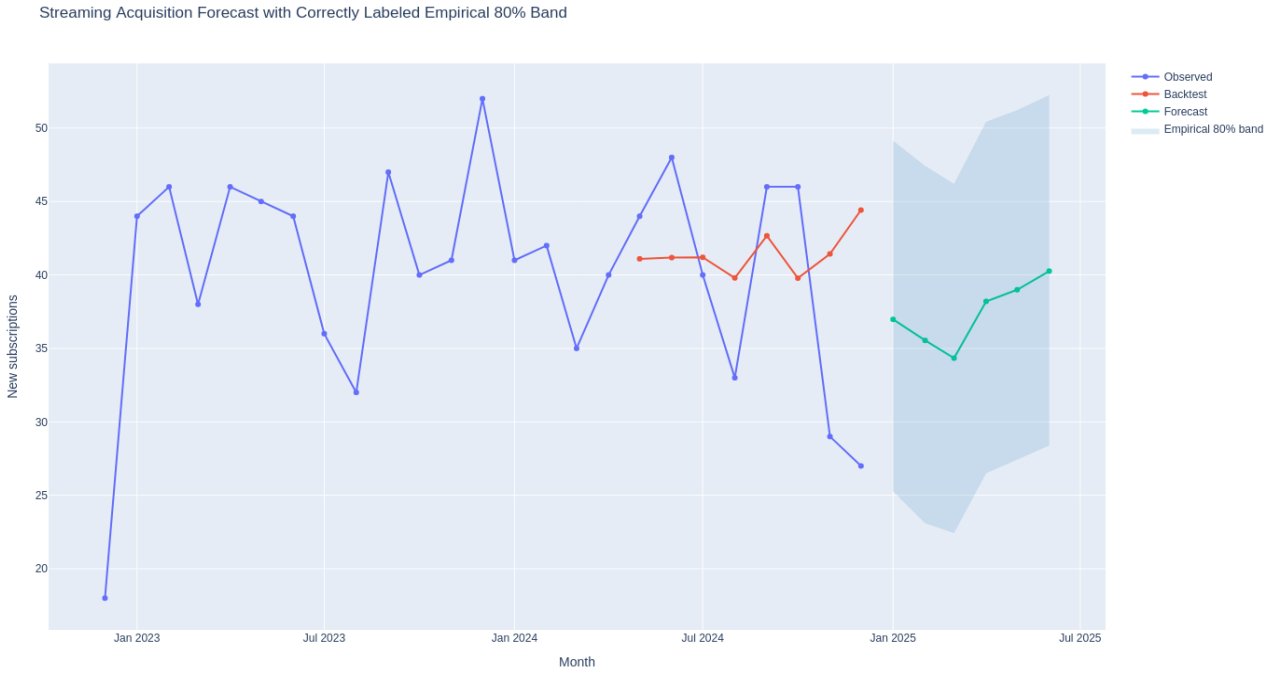
Retention Hazard Curves V2



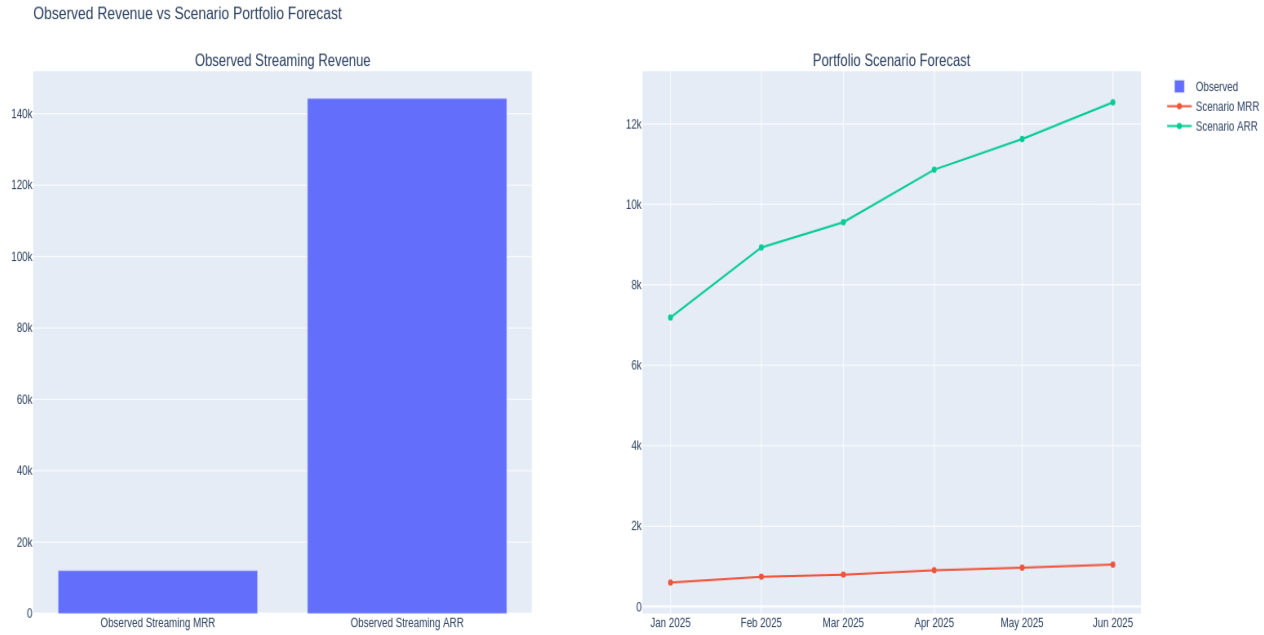
Calibration By Contract V2



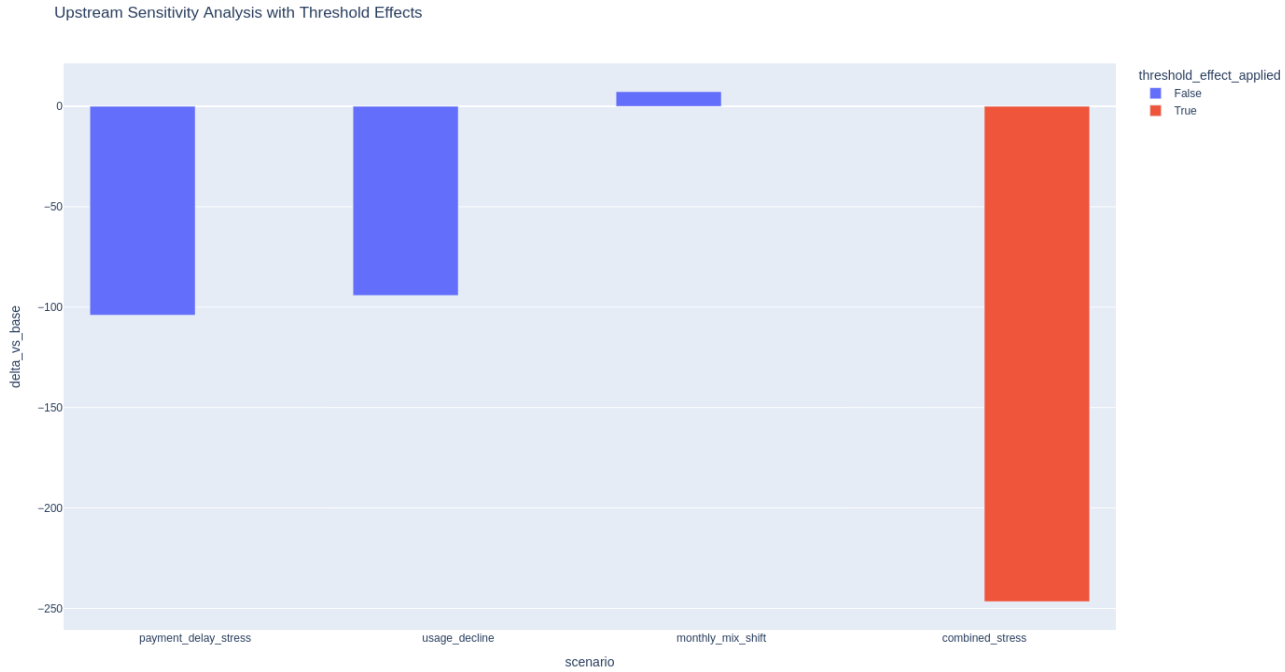
Streaming Acquisition Forecast V2



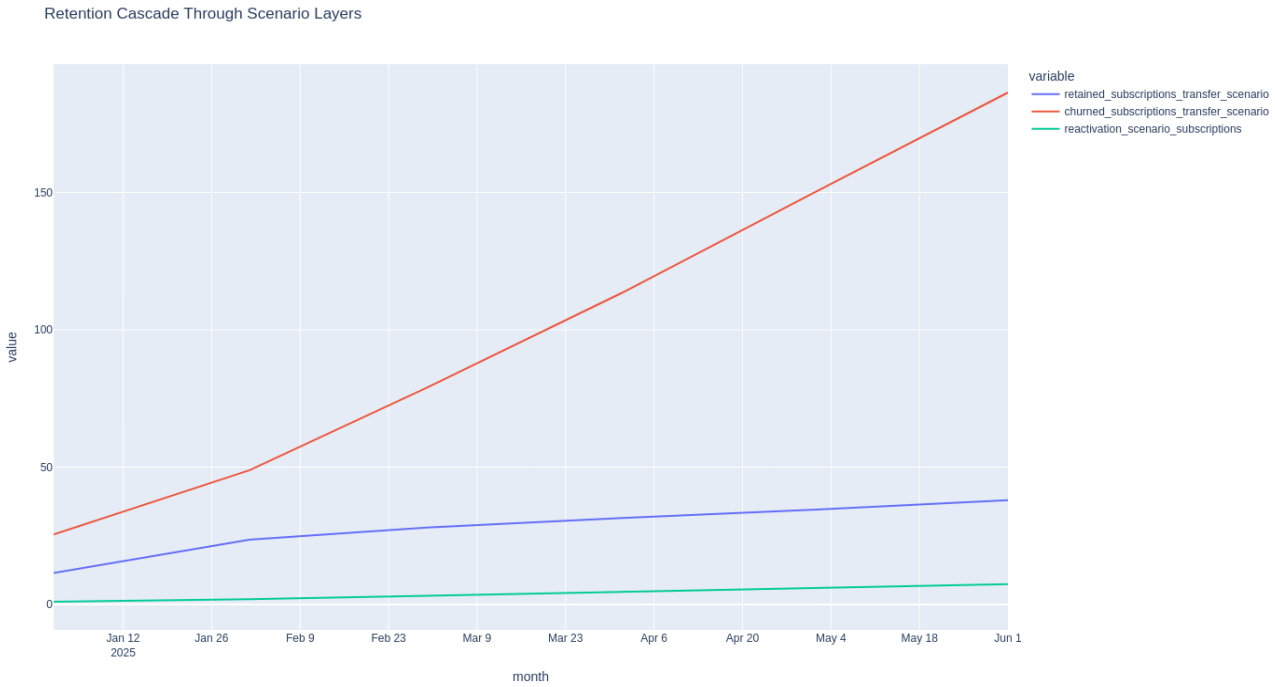
Observed Vs Scenario Revenue V2



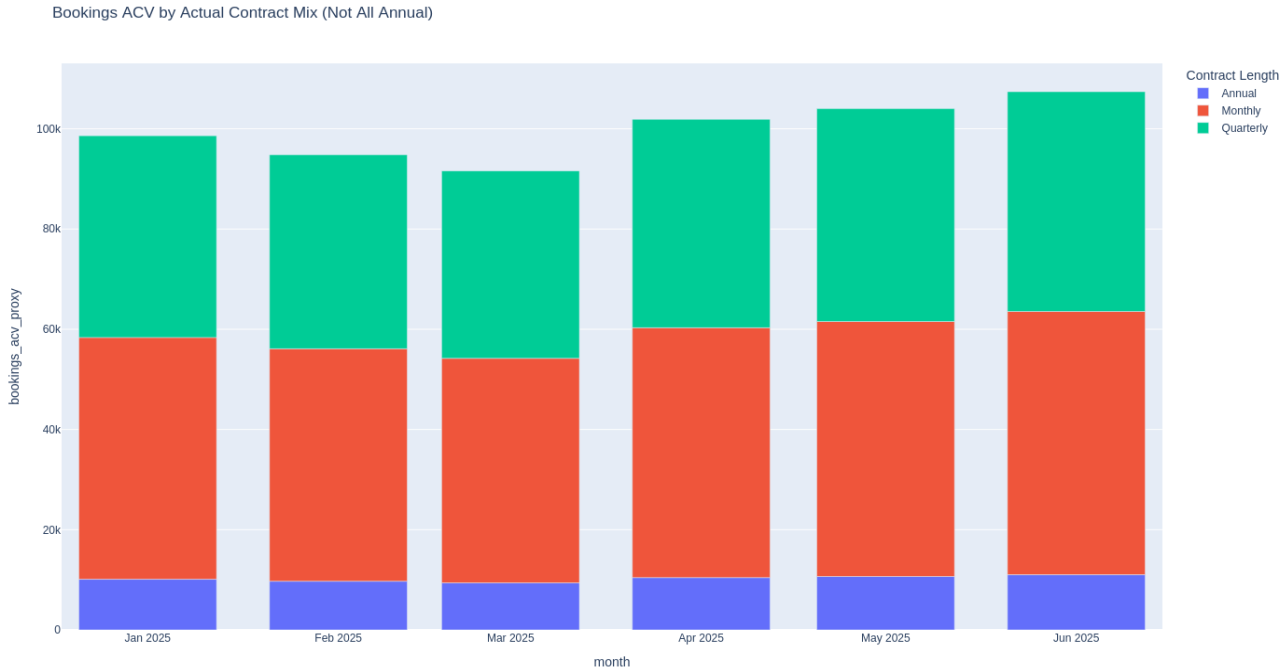
Upstream Sensitivity Analysis V2



Retention Cascade Scenarios V2



Bookings By Contract Mix V2

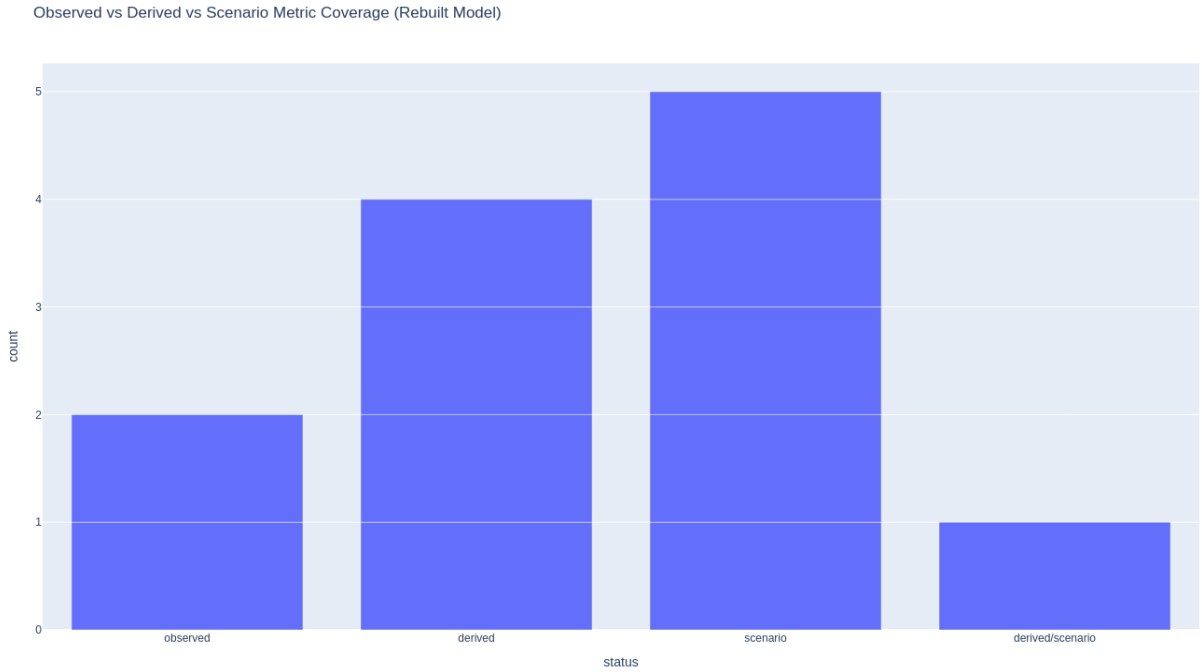


Anomaly Alerts V2

Anomaly Detection Alerts

alert	triggered
terminal_trajectory_alert	true
unit_economics_failure_alert	true
constant_expansion_alert	false
monthly_contract_artifact_alert	false
economically_nonsensical_alert	false

Metric Status Coverage V2


Observed vs Derived vs Scenario Metric Coverage (Rebuilt Model)
count
0
1
2
3
4
5
observed
derived
scenario
derived/scenario
status